



User Guide for Neon Download Tool

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Revision History

Version	Date	Comments
1.0	02-20-2026	Initial version
2.0	03-12-2026	Enabled Reset button and update on Section 5.5

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Terms & Definitions

ELF	Executable and Linkable Format
EVK	Evaluation Kit
FTDI	Future Technology Devices International
GUI	Graphical User Interface
SDK	Software Development Kit
SWD	Serial Wire Debug

1. Overview

This document describes the use of the Download Tool provided with Neon SDK. The Download Tool is provided as binaries for Windows and Linux Operating Systems.

Download Tool is a GUI-based application designed to perform flashing operations on Neon and monitor console outputs from Neon.

The application `Neon_DownloadTool_Windows.exe` is intended for the Windows platform, while `Neon_DownloadTool_Linux` is designed for the Linux operating system. This document provides GUI usage instructions based on the Linux version (`Neon_DownloadTool_Linux`) of the application. The interface and functionality are similar for the Windows version.

2. Prerequisites

The Download Tool is supported on Windows 10 (or newer) and Ubuntu 22.04 (or newer) operating systems.

2.1 Device setup

Neon RK Board:

Setup requires:

- Neon RK board
- C-Type cable

The connections are illustrated in Figure 1. Connect the Neon RK board to the PC using C-Type Cable.

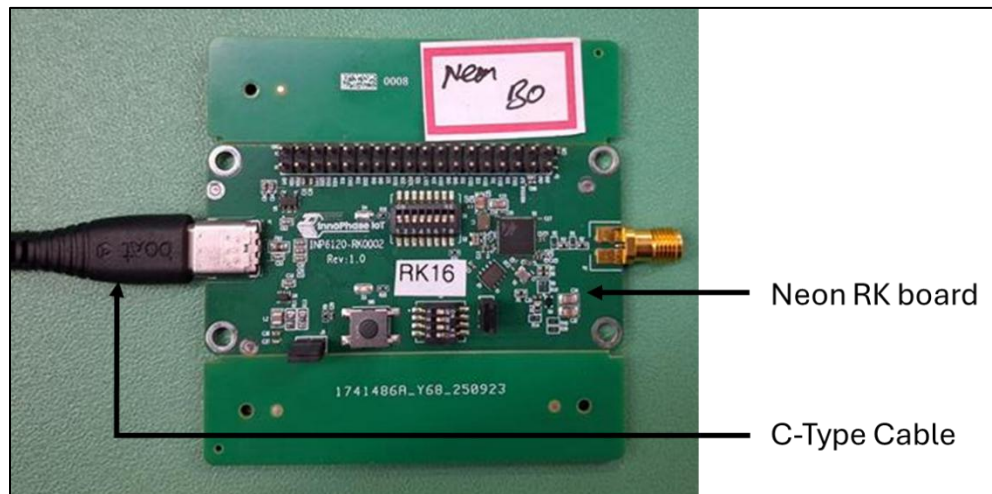


Figure 1: Connection for Neon RK board

Note: The FTDI interface is built into the RK board; therefore, an external programmer board is not required.

Neon PP Board:

Device setup requires:

- Neon PP board
- INP3000 programmer board
- SWD cable (to connect the Neon PP to the INP3000 board)
- Customized cable (to connect Neon console to the INP3000 board)
- Power supply cable (C-Type) to Neon PP Board
- Micro USB cable (to connect the INP3000 board to the PC)

The connections are illustrated in Figure 2.

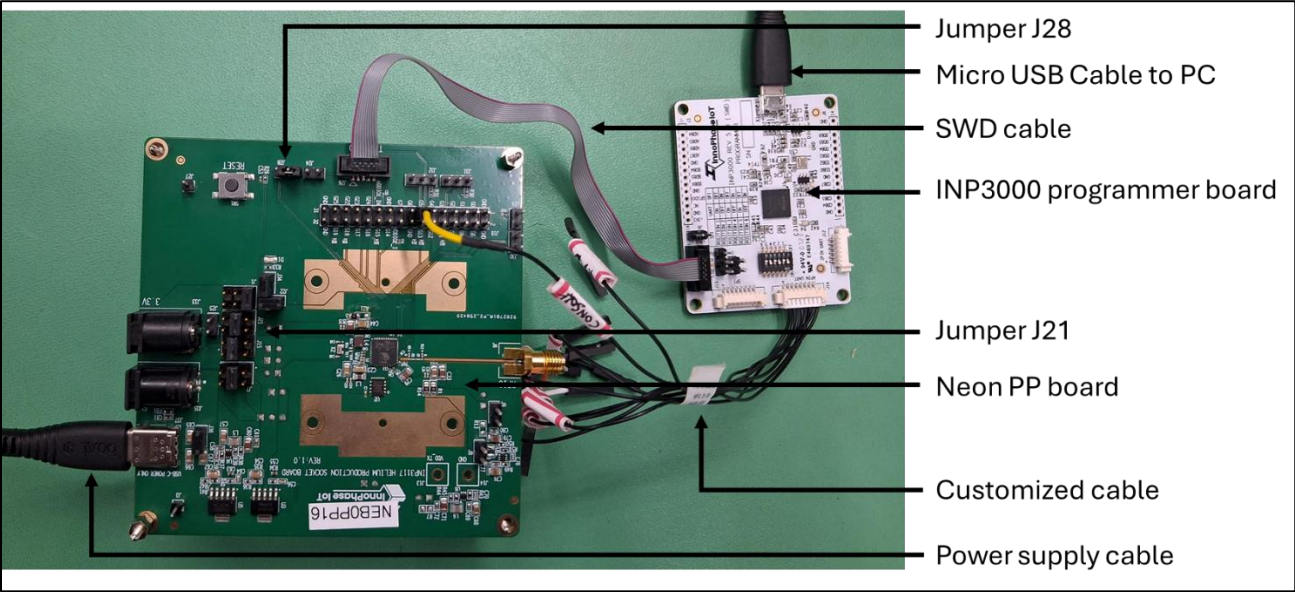


Figure 2: INP3000 SWD Programmer Board with SWD interface to Neon PP board

Step procedure to connect the device setup to PC:

1. Connect the IN3000 programmer board to Neon PP board as shown in Table 1.

Cable connection configuration		
Cable	INP3000 programmer board	Neon PP board
Customized cable	J14	Console pin -> GPIO 5 GND -> GND
SWD cable	J6	J26
3.3v Jumper	-	J21 -> 3 and 4 Shorted
J28 Jumper	-	J28 Shorted for reset through SWD

Table 1: Cable connection configuration between Neon PP board and IN3000 programmer board

2. Connect the INP3000 programmer board to the PC.
3. Power the Neon PP board using power cable.

2.2 Using Download Tool in Linux

There are two ways in which the `Neon_DownloadTool_Linux` can be used in Linux:

1. Run the `Neon_DownloadTool_Linux` tool from terminal with `sudo` command. For example:

```
sudo /sdk_x.y.z/Tools/Neon_DownloadTool_Linux
```

Note: x, y and z in `sdk_x.y.z` refers to the SDK release version. Replace x, y and z with the appropriate SDK release version.

2. Double click on `Neon_DownloadTool_Linux`.

Add udev rules and folder access permission to enable double click feature. Execute the following steps:

Step 1: Create `Libusb_innophase.rules` file in `/etc/udev/rules.d` directory.

Step 2: Add the following rules to `Libusb_innophase.rules` file.

Libusb_innophase.rules:

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="0403", ATTRS{idProduct}=="6011",  
  
GROUP="users", MODE="0666"
```

In case of `Permission Denied` error, execute the following step (Step 3) to extend folder access permission.

Step 3: Run the following command in the terminal to extend permissions to the selected folder and its files.

```
sudo chmod -R a+rw-rwx /Path/to/sdk_folder
```

2.3 Microsoft Visual C++ Redistributable Software Package

Microsoft Visual C++ Redistributable software package is a prerequisite for Windows platform to run the application `Neon_DownloadTool_Windows.exe` successfully. In case this software package is not installed on the Windows platform, application will not launch, leading to a fatal error message as shown in Figure 3. In such a scenario, install the Microsoft Visual C++ Redistributable software package and relaunch the application.

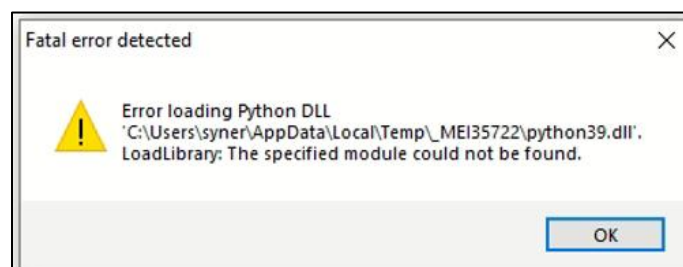


Figure 3: Error message for missing Microsoft Visual C++ Redistributable Software Package

2.4 Driver Installation for Windows OS

On Windows OS, libusbK driver needs to be installed to communicate and control the Neon via the INP3000 Programmer board or FTDI Interface. The Download Tool communicates with Neon through INP3000 Programmer board or FTDI interface using this driver.

Download Tool comes with an option of One-Click Installation of libusbk driver. In case the driver is not installed, the tool will ask for user confirmation to install this driver. If the user selects `yes`, various User Account Control authentication screens will appear to complete the driver installation (as shown in Figure 6).

Note:

1. In case of any other unwanted libusbk drivers that are already installed, tool will automatically uninstall the unwanted drivers. This action needs User Account Control authentication screens shown in Figure 7, in addition to Figure 6. Each unwanted drives will require a separate User Account Control authentication for uninstallation.
2. INP3000 Board may get detected under any already installed device driver (using libusbk driver). In this case, the tool will follow the procedure of One-Click Installation of libusbk driver (as shown in Figure 6), and might get completed with a pop-up message “Driver Installation Failed”. This will result in any one of the following two conditions:
 - a. Device found: EVK serial number field in the Download Tool GUI is populated with appropriated INP3000 or FTDI interface serial number. In this case, ignore the error message “Driver Installation Failed” and continue using the tool. To avoid this from happening repeatedly, update the driver for IN3000 Board to “InnoPhase Evaluation Board” driver, from device manager (as shown in Figure 4). Then, ensure the INP3000 Board is detected under “libusbK USB Devices” in device manager (as shown in Figure 5).



Figure 4: Update device driver for INP3000 Board

- b. No device found: Manually install the driver using instructions from section: 2.5 Manual installation instructions for libusbK driver and ensure that INP3000 Board is detected under libusbK USB Devices driver (as shown in Figure 5).

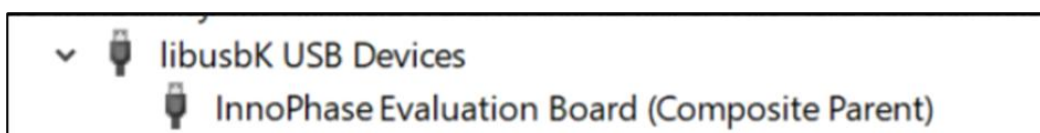


Figure 5: INP3000 board under “libusbk USB Devices” driver

3. Even after successful installation of the driver, there might be possibility of the device not being identified for the first time. In this case, close the tool and re-open it. The user is notified of the same through a pop-up message: “No Device Connected. Please close and reopen the Tool.”.

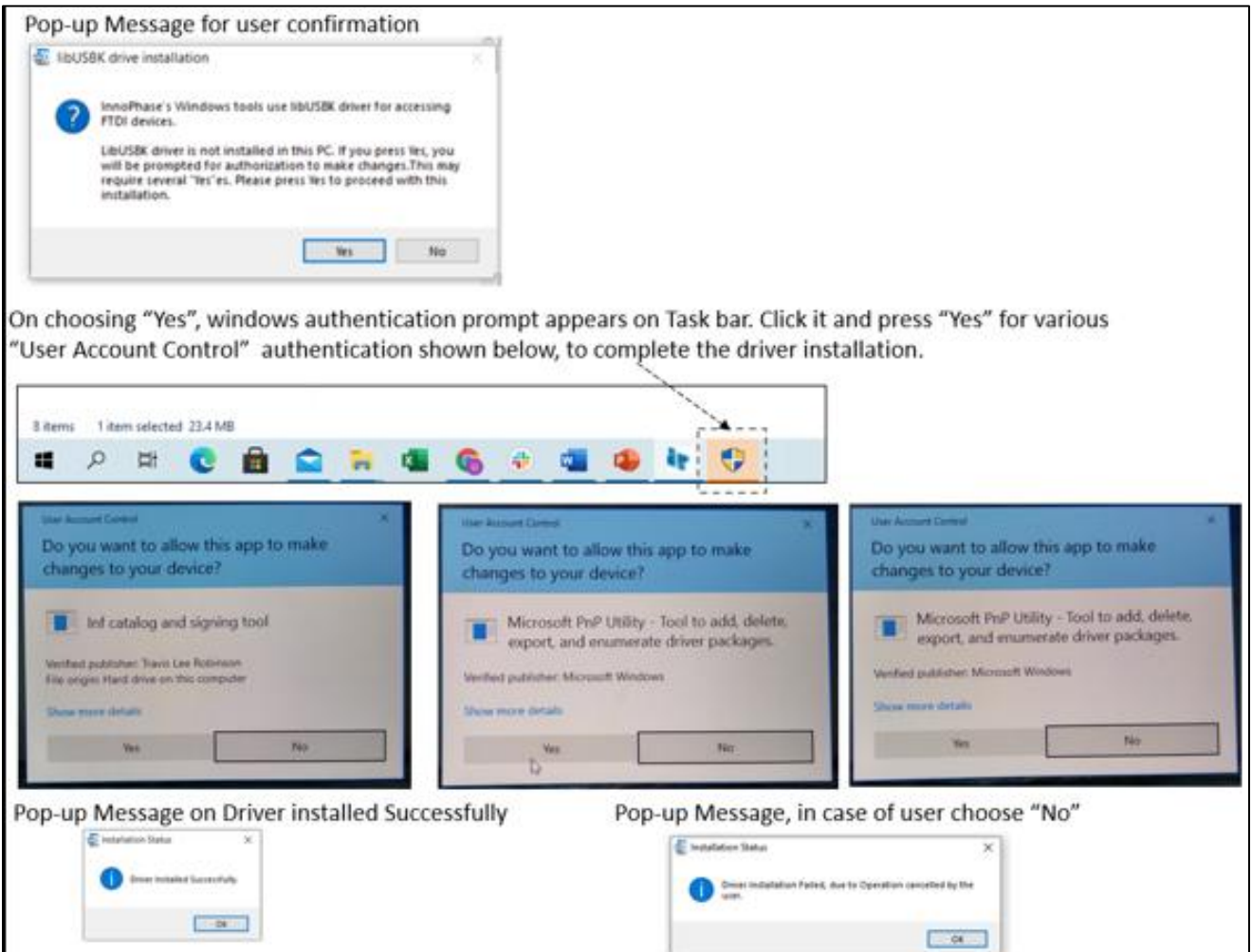


Figure 6: User Account Control authentication to complete driver installation

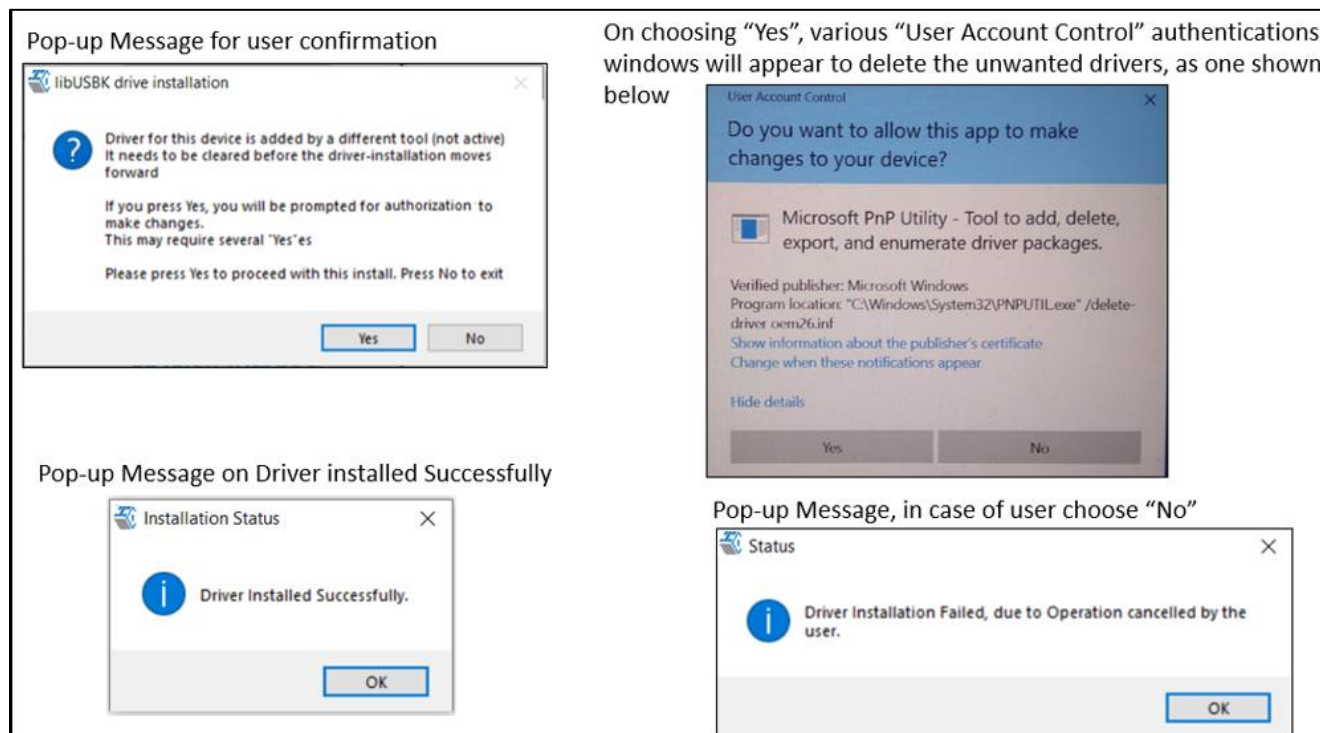


Figure 7: User Account Control authentication to delete the unwanted Libusbk drivers

In case the driver installation using the Download Tool is not successful, the user can manually install the driver by following the instructions provided in section: 2.5 Manual installation instructions for libusbK driver.

2.5 Manual installation instructions for libusbK driver

Download the free software Zadig, available here: - <https://zadig.akeo.ie/>. Connect your Windows PC or Windows Laptop to the evaluation board using the provided USB cable.

1. Now, open Zadig
2. Click on Options
3. Check List All Devices
4. Uncheck Ignore Hubs or Composite Parents as shown in Figure 8.
5. From the dropdown menu, select the INP3000 or Innophase board (composite Parent) device, which can be identified by the name string 'Innophase'.

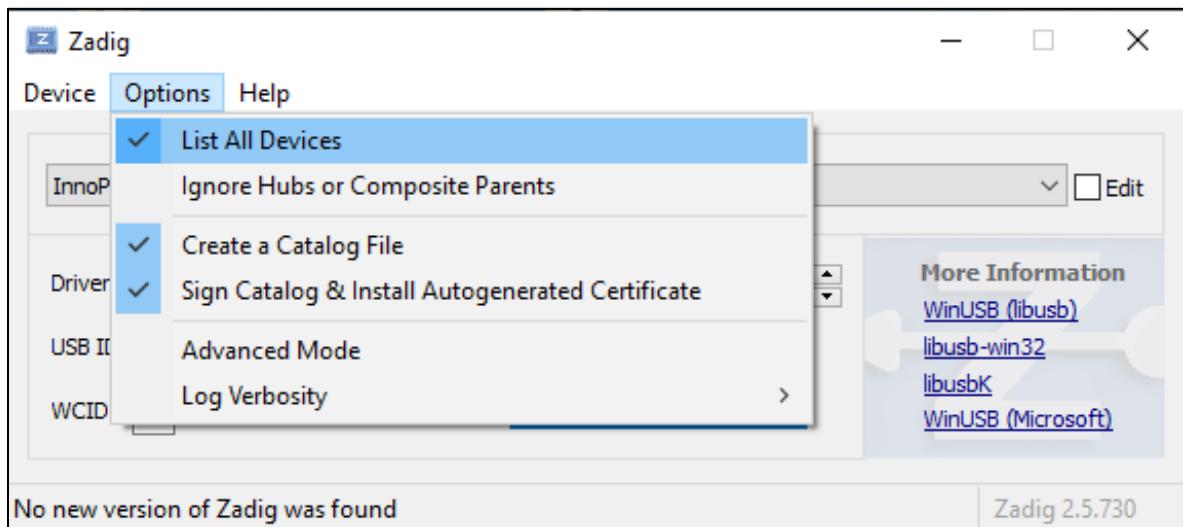


Figure 8: Listing devices in Zadig

To establish communication with Neon via the FTDI device on the INP3000 or Innophase Board, the USB driver must be libusbK. In case the current driver is not libusbK, use the drop-down menu to select libusbK and click on Replace Driver which will update the drivers to libusbK.

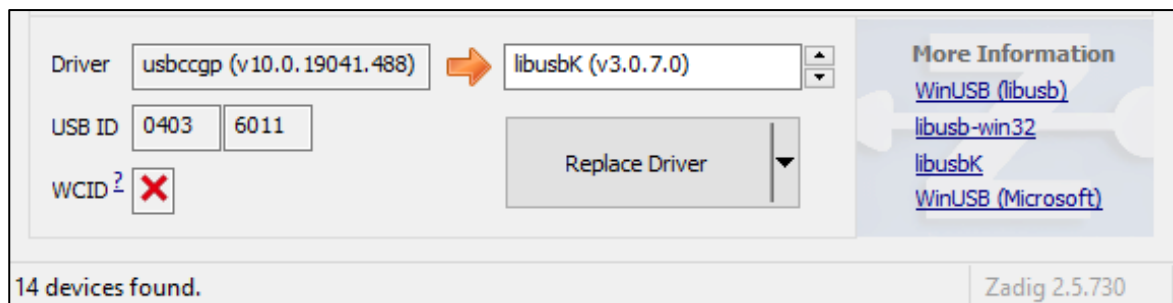


Figure 9: Updating INP3000 or Innophase board driver to libusbK

2.6 OpenOCD Installation

OpenOCD software should be installed for programming using SWD interface.

On the Linux OS:

Install using the following command:

```
sudo apt install openocd -y
```

Note: Ensure that OpenOCD version is 0.11.0 or higher.

On Windows OS:

1. Download the standalone Prebuilt Windows Binaries from:
<https://github.com/xpack-dev-tools/openocd-xpack/releases/download/v0.10.0-15/xpack-openocd-0.10.0-15-win32-x64.zip>
2. Extract the downloaded archive.
3. Add the full path of bin directory (e.g., C:\path\to\xpack-openocd-0.10.0-15-win32-x64\xpack-openocd-0.10.0-15\bin) to the **System Environment Variables -> PATH**.

Notes:

1. OpenOCD is managed automatically by the Download tool.
2. This setup is required only for the boot script.
3. Ensure that OpenOCD version is v0.10.0-15 or lower

2.7 GDB-Multiarch Installation

GDB-Multiarch software should be installed for recover device from double fault issue.

Note: As mentioned in Table 1, jumper J28 need to be shorted to recover from double fault issue.

On Linux OS:

Install using the following command:

```
sudo apt install gdb-multiarch -y
```

On Windows OS:

4. Download the standalone Prebuilt Windows Binaries from the Grumpycoder server:
<https://static.grumpycoder.net/pixel/gdb-multiarch-windows>
5. Select the latest version (recommended).
At the time of development, the latest was **gdb-multiarch-16.3.zip**
6. Unzip the archive.
7. Add the bin folder path (e.g., C:\path\to\gdb-multiarch-16.3\bin) to the **System Environment Variables -> PATH**.

2.8 libxcb-cursor0 Installation

In case of encounter the following issue while launching Neon_DownloadTool_Linux on a Linux system,

```
hostnamectl: /tmp/_MEIsn9hMD/libselinux.so.1: no version information available (required
by /lib/systemd/libsystemd-shared-249.so)

qt.qpa.plugin: From 6.5.0, xcb-cursor0 or libxcb-cursor0 is needed to load the Qt xcb
platform plugin.

qt.qpa.plugin: Could not load the Qt platform plugin "xcb" in "" even though it was
found.

This application failed to start because no Qt platform plugin could be initialized.
Reinstalling the application may fix this problem.

Available platform plugins are: minimal, vkkhrdisplay, linuxfb, wayland, xcb, vnc,
wayland-egl, eglfs, offscreen, minimalegl.

Aborted
```

install the required support package by running the command below:

```
sudo apt install libxcb-cursor0
```

3. Download Tool GUI Overview

When the Download Tool application is launched, the graphical user interface (GUI) appears as shown in Figure 10. The upper section includes input controls, and the lower section provides a console window to display Neon console logs (as illustrated in Figure 11).

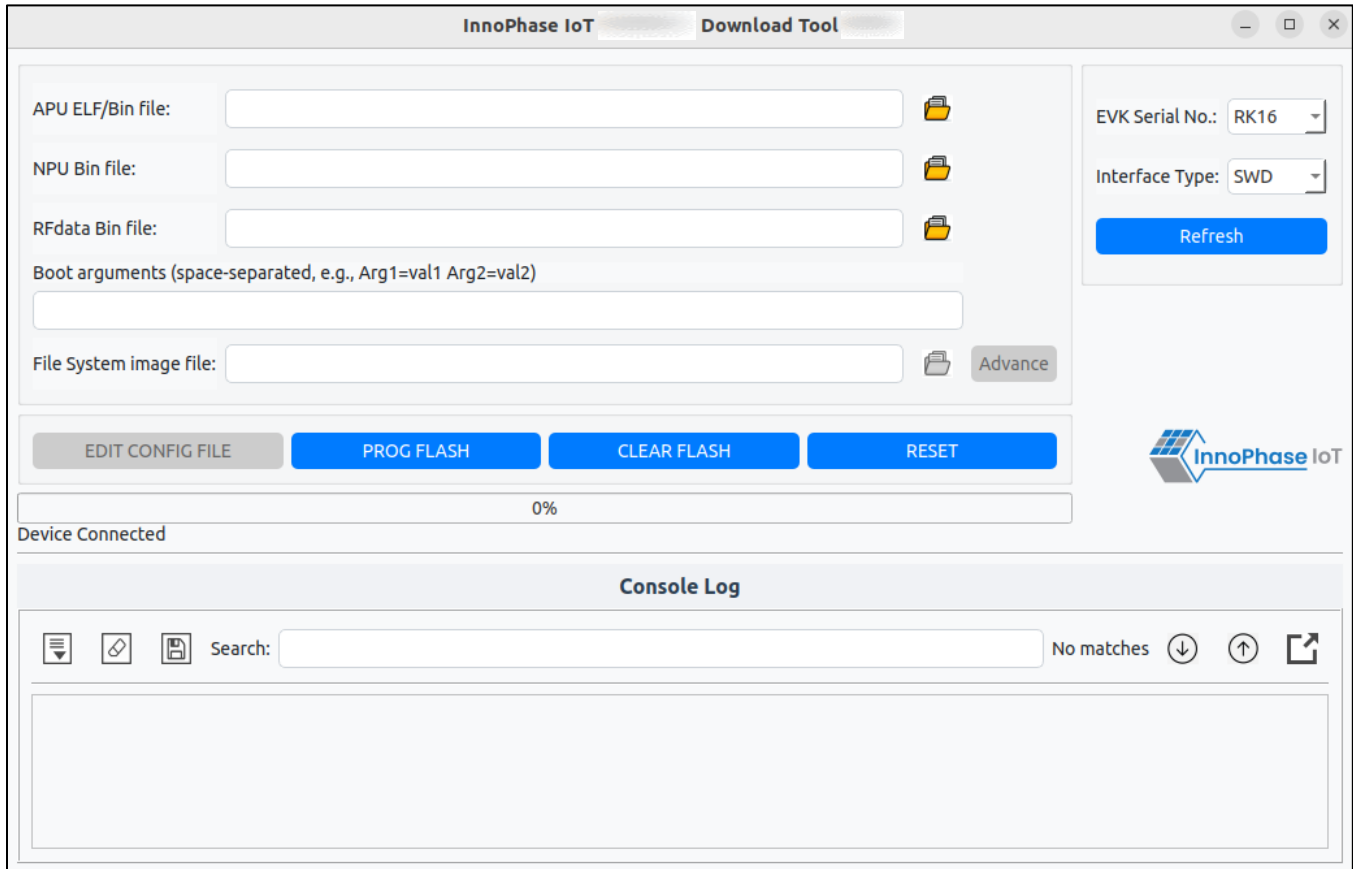


Figure 10: Download Tool GUI



Figure 11: Download Tool GUI – Console log Window

The console window has the following icons:

1. **Auto Scroll** : Enables scrolling of console content till the end (default mode).
2. **Pause Scroll** : Turns OFF Auto Scroll mode.
3. **Clear Console** : Clears console window content.
4. **Save Logs** : Opens a file dialog with Console_Output.log as the default file name to save the logs.
5. **Stop Save Logs** : Stops saving console logs to the file. This icon appears after Save Logs is started successfully.
6. **Search option** : Allows users to search for a specific string within the console logs. It includes Search Up and Search Down functionalities for navigating through the matches.
7. **Pop Out** : Pops out the console window separate from the GUI window.
8. **Pop In** : Embeds the console and GUI window together.

4. Download using Command Line

The Download GUI tool internally uses a boot script to download applications to the Neon device. These backend scripts are packaged with the SDK (location `/sdk_x.y.z/Tools/scripts/`) and can also be used directly via the command line.

Steps to Use the Boot Script:

Prerequisite (First-time setup):

Install the required Python libraries:

```
pip install -r requirements.txt
```

Clear Flash:

To clear the flash memory before programming:

```
python3 boot.py --clear_flash
```

Program Flash:

To program an application image (elf or bin format):

```
python3 boot.py --app_elf ../apps/hello_world/bin/hello_world.elf
```

Reset Device:

Reset device to restart the application:

```
python3 boot.py --reset_device
```

Notes:

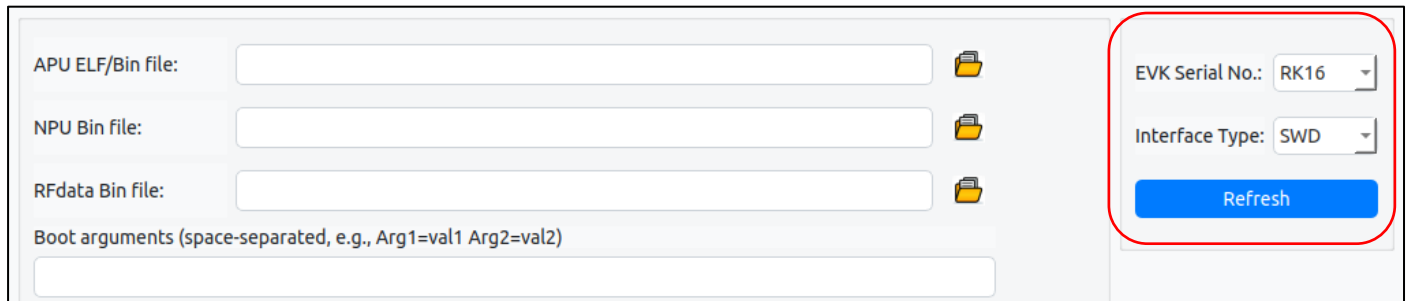
1. w, x, y, and z in `sdk_x.y.z` refer to the SDK release version. Replace them with the appropriate version (e.g., `sdk_1.0.0`).
2. Run all the above commands from within the `/sdk_x.y.z/scripts/` directory.
3. Command-line Arguments:
 - `--app_elf`: Specifies the path to the ELF or BIN file of the application to be programmed (e.g., `../apps/hello_world/bin/hello_world.elf`)
 - `--serial`: Specifies the serial number of the target device (e.g., `30005015`).
This argument is optional and is required only when multiple devices are connected.
4. For Windows OS: use `python` instead of `python3` in the command.

5. Download Tool Usage

5.1 Target serial Number and Interface

Download Tool supports programming using UART and SWD interface. Select the appropriate interface as per the connection from the `Interface Type` drop-down (SWD being the default selection).

Connected INP3000 device appear in the EVK serial number drop-down and the appropriate EVK can be selected.



The screenshot shows the 'Boot target - EVK selection' interface. On the left, there are four input fields for file selection: 'APU ELF/Bin file:', 'NPU Bin file:', 'RFdata Bin file:', and 'Boot arguments (space-separated, e.g., Arg1=val1 Arg2=val2)'. Each file field has a folder icon to its right. On the right side, there is a red-bordered box containing the 'EVK Serial No.' dropdown menu (set to 'RK16'), the 'Interface Type' dropdown menu (set to 'SWD'), and a blue 'Refresh' button.

Figure 12: Boot target - EVK selection

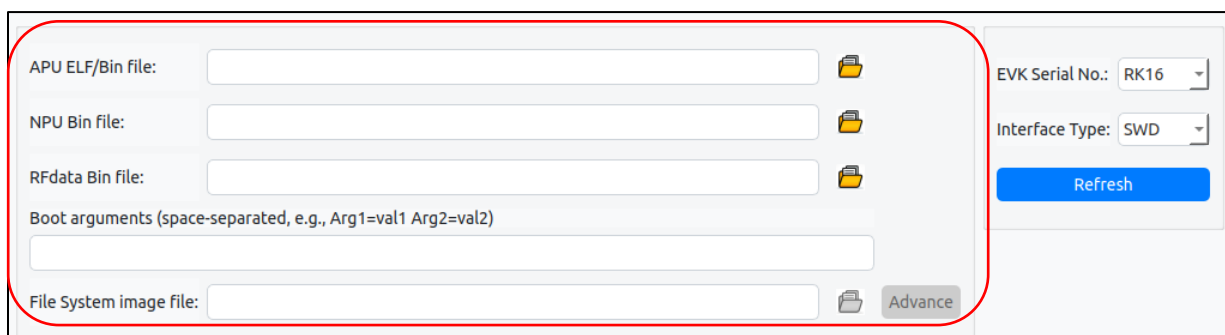
Note:

1. Refresh Button: Scans and updates the status of connected devices to reflect any recent changes.
2. At present, programming is supported only via the SWD interface. UART programming functionality is pending implementation.

5.2 Program Flash inputs

Selecting an Appropriate APU ELF, NPU Bin, RFdata Bin, Boot arguments and File System Image File File:

1. APU ELF/Bin File: Select the APU application ELF or BIN file to be downloaded to the target device.
2. NPU Bin File: Select the NPU application BIN file to be downloaded to the target device.



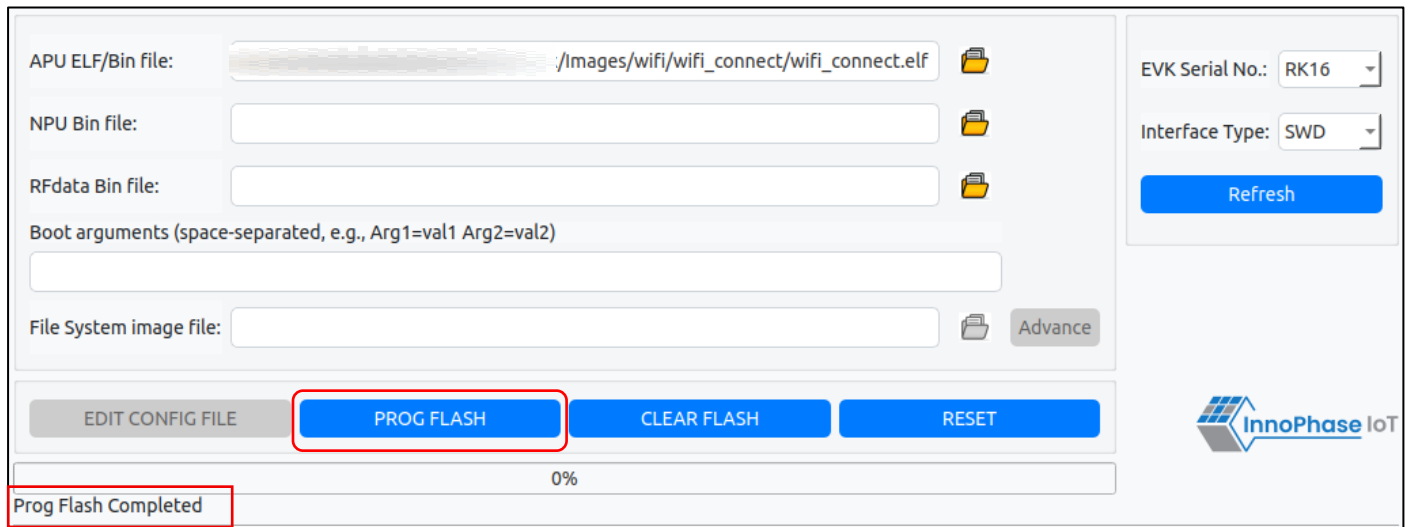
The screenshot shows the 'Program Flash Inputs' interface. It includes the same file selection fields as Figure 12, plus a 'File System image file:' field at the bottom with a folder icon. A red-bordered box highlights the file selection fields. To the right, the 'EVK Serial No.' and 'Interface Type' dropdowns are visible, along with the 'Refresh' button and a grey 'Advance' button at the bottom right.

Figure 13: Program Flash Inputs

3. RFdata Bin file: Select the RF data calibration application BIN file to be downloaded to the target device.
4. Boot Arguments: Provide the required NPU boot arguments in space-separated format (e.g., Arg1=val1 Arg2=val2).
5. File System Image File (currently disabled): Select the file system image needs to be downloaded to the DATA partition during the PROG FLASH operation.

5.3 Programming to Flash

- Provide the inputs for Program flash as described in section 5.2 Program Flash inputs, and click on PROG Flash, as shown in Figure 14 & Figure 15.
- Console output will be shown on “Console Log” window, as shown in Figure 16.
- Once program flash is completed, status message shows “Prog Flash Completed”, as highlighted in Figure 14.
- The console output of application flashed will be captured on “Console Log” window, as highlighted in Figure 16.



APU ELF/Bin file: 

NPU Bin file: 

RFdata Bin file: 

Boot arguments (space-separated, e.g., Arg1=val1 Arg2=val2)

File System image file:  Advance

EVK Serial No.:

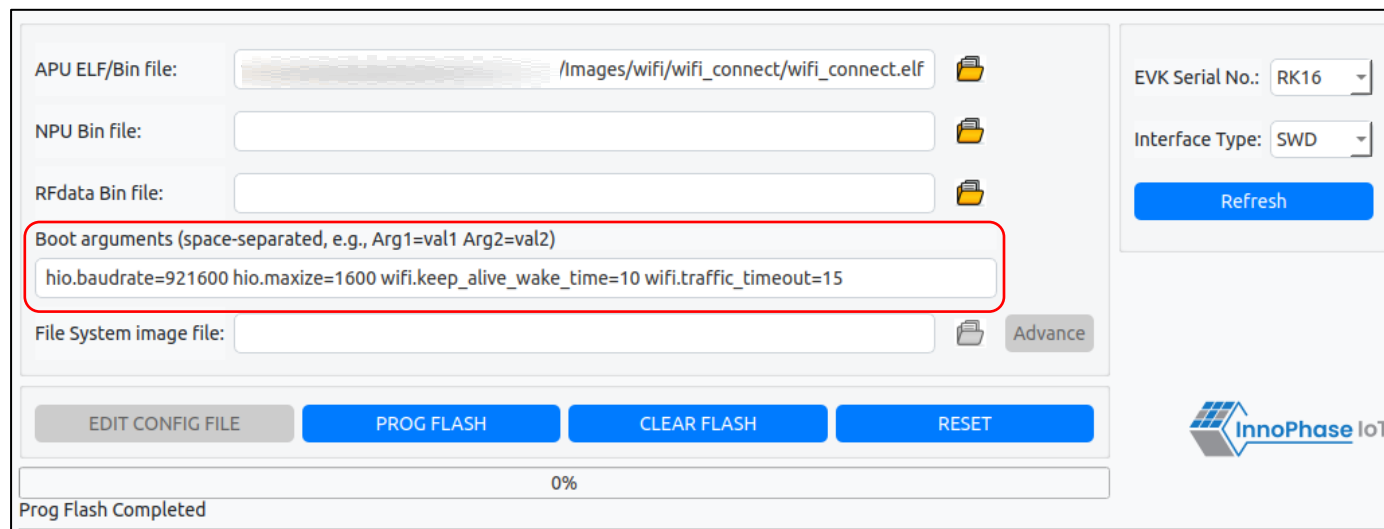
Interface Type:

Refresh

EDIT CONFIG FILE **PROG FLASH** CLEAR FLASH RESET

Prog Flash Completed 0%

Figure 14: PROG Flash



APU ELF/Bin file:

NPU Bin file:

RFdata Bin file:

Boot arguments (space-separated, e.g., Arg1=val1 Arg2=val2)

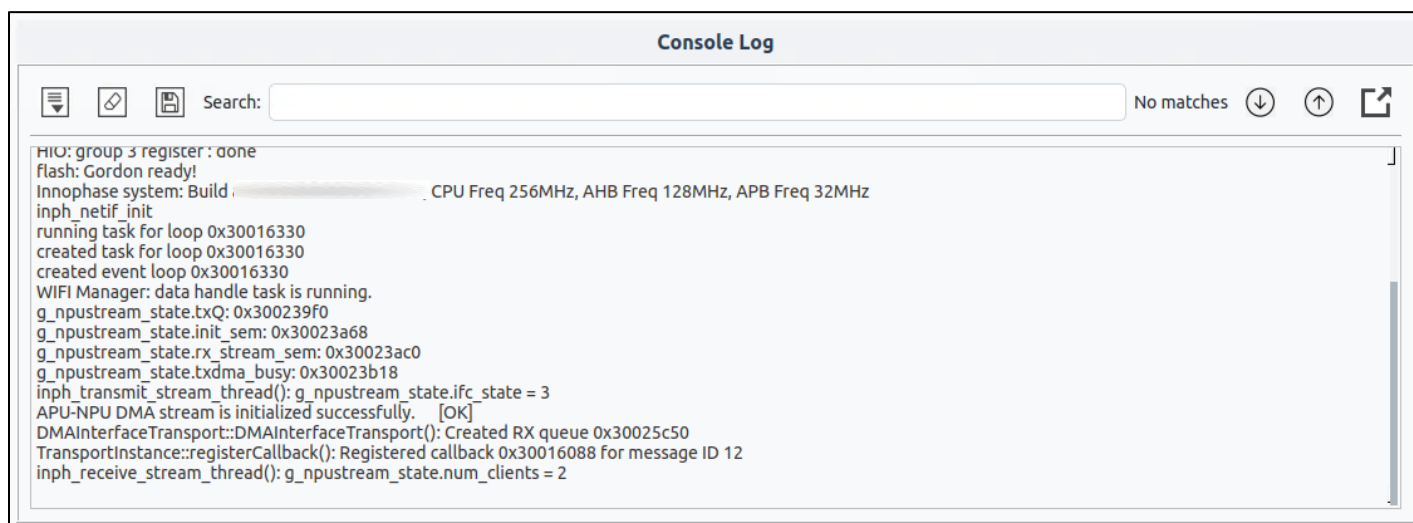
File System image file:

EVK Serial No.:

Interface Type:

Prog Flash Completed 0%

Figure 15: PROG Flash with Boot Arguments input



Console Log

Search: No matches

HIO: group 3 register: done
flash: Gordon ready!
Innophase system: Build ... CPU Freq 256MHz, AHB Freq 128MHz, APB Freq 32MHz
inph_netif_init
running task for loop 0x30016330
created task for loop 0x30016330
created event loop 0x30016330
WIFI Manager: data handle task is running.
g_npustream_state.txQ: 0x300239f0
g_npustream_state.init_sem: 0x30023a68
g_npustream_state.rx_stream_sem: 0x30023ac0
g_npustream_state.txdma_busy: 0x30023b18
inph_transmit_stream_thread(): g_npustream_state.ifc_state = 3
APU-NPU DMA stream is initialized successfully. [OK]
DMAInterfaceTransport::DMAInterfaceTransport(): Created RX queue 0x30025c50
TransportInstance::registerCallback(): Registered callback 0x30016088 for message ID 12
inph_receive_stream_thread(): g_npustream_state.num_clients = 2

Figure 16: PROG Flash - Console Output

5.4 Clear Flash

- Use `CLEAR FLASH` to clear the application on Neon flash, as shown in Figure 17.
- Console output will be shown on “Console Log” window, as shown in Figure 18.
- Once Clear flash is completed, status message shows “Clear Flash Completed”, as highlighted in Figure 17.

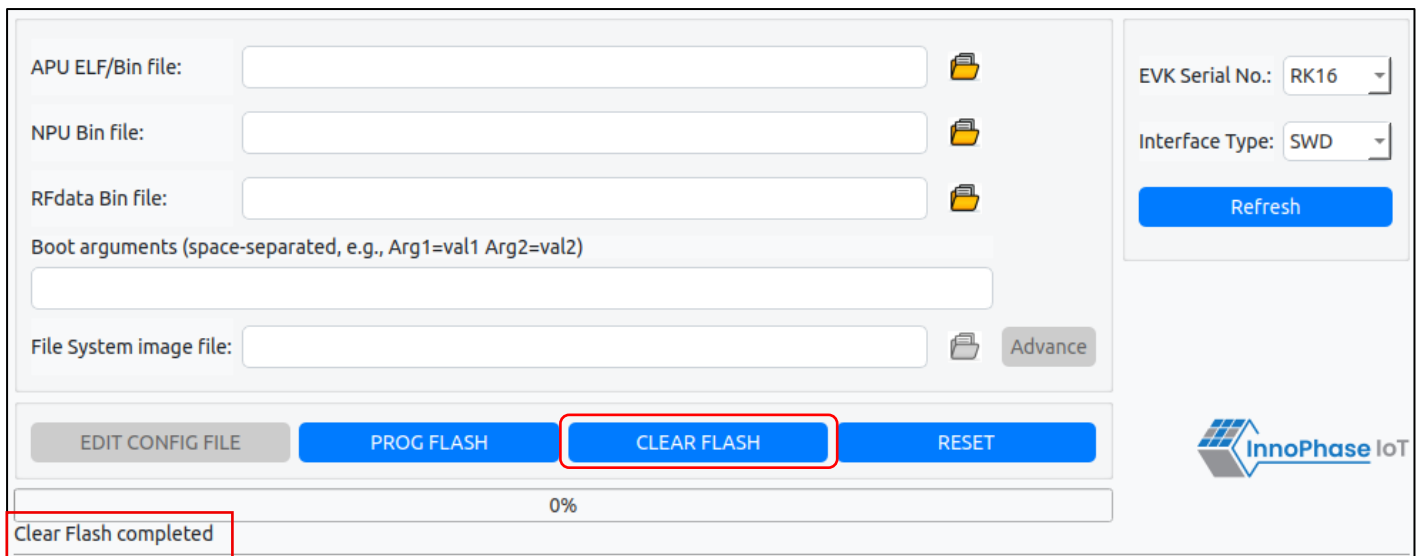


Figure 17: CLEAR FLASH

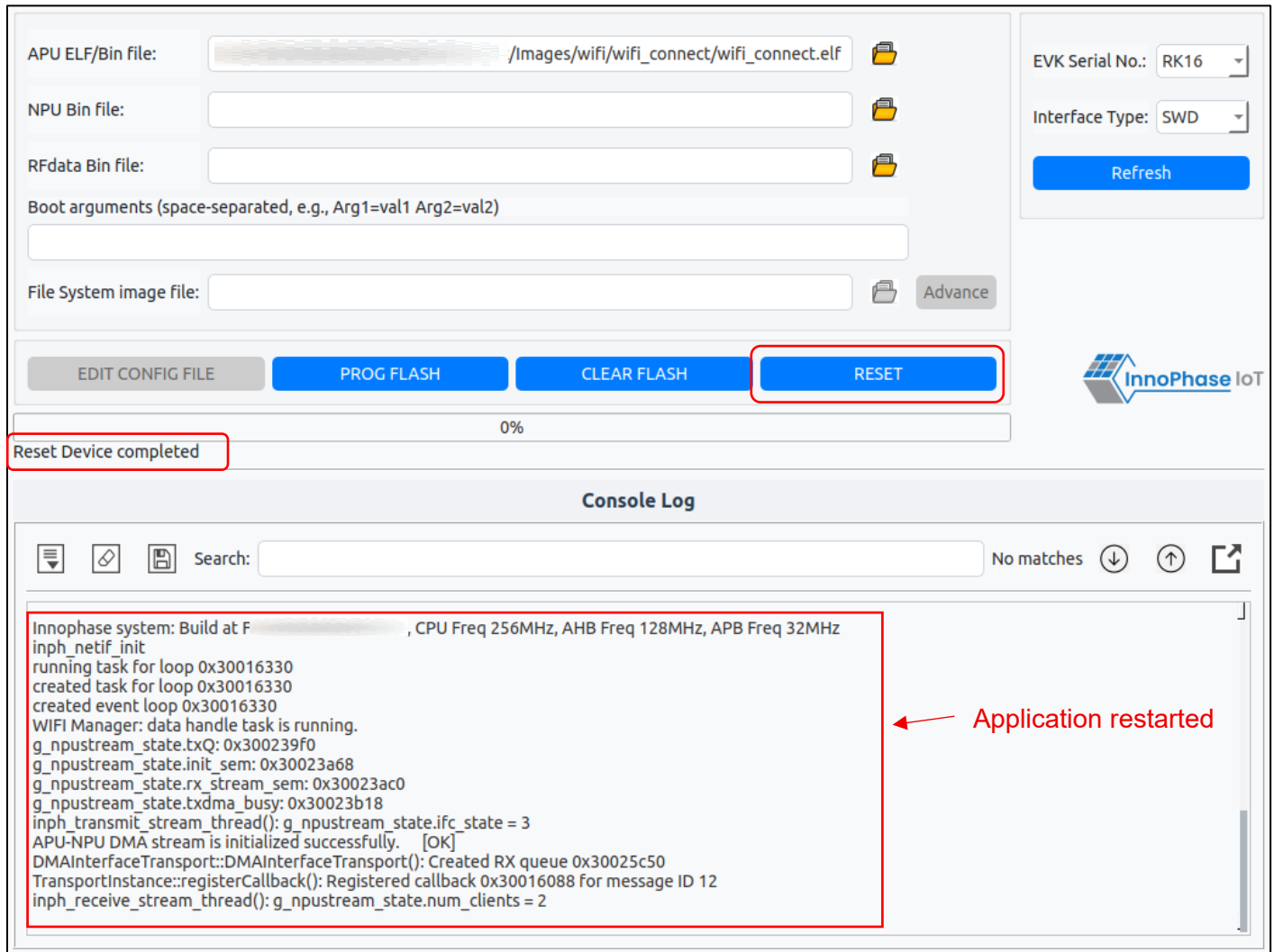


Figure 18: CLEAR FLASH - Console Output

Note: During the Clear Flash operation, the console output message “Flash Sector Erase Done” may not always appear. However, this does not indicate a failure—the flash is typically erased successfully. To confirm, perform a power reset on the device.

5.5 Reset

Click **RESET** button to restart the application stored in flash memory. The running application will stop and restart automatically.



The screenshot displays the Neon Download Tool interface. On the left, there are input fields for 'APU ELF/Bin file:', 'NPU Bin file:', 'RFdata Bin file:', 'Boot arguments (space-separated, e.g., Arg1=val1 Arg2=val2)', and 'File System image file:'. To the right, there are dropdown menus for 'EVK Serial No.: RK16' and 'Interface Type: SWD', along with a 'Refresh' button. Below these are buttons for 'EDIT CONFIG FILE', 'PROG FLASH', 'CLEAR FLASH', and 'RESET'. The 'RESET' button is highlighted with a red rectangle. Below the buttons is a progress bar showing '0%' and a status message 'Reset Device completed' in a red-bordered box. The bottom section is the 'Console Log', which contains a search bar and a list of system logs. A red-bordered box highlights the log messages, and a red arrow points to it with the text 'Application restarted'.

APU ELF/Bin file: 

NPU Bin file: 

RFdata Bin file: 

Boot arguments (space-separated, e.g., Arg1=val1 Arg2=val2)

File System image file:  **Advance**

RESET

0%

Reset Device completed

Console Log

Search: No matches   

Innophase system: Build at F , CPU Freq 256MHz, AHB Freq 128MHz, APB Freq 32MHz
inph_netif_init
running task for loop 0x30016330
created task for loop 0x30016330
created event loop 0x30016330
WIFI Manager: data handle task is running.
g_npustream_state.txQ: 0x300239f0
g_npustream_state.init_sem: 0x30023a68
g_npustream_state.rx_stream_sem: 0x30023ac0
g_npustream_state.txdma_busy: 0x30023b18
inph_transmit_stream_thread(): g_npustream_state.ifc_state = 3
APU-NPU DMA stream is initialized successfully. [OK]
DMAInterfaceTransport::DMAInterfaceTransport(): Created RX queue 0x30025c50
TransportInstance::registerCallback(): Registered callback 0x30016088 for message ID 12
inph_receive_stream_thread(): g_npustream_state.num_clients = 2

Application restarted

Figure 19: Reset

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